A SPLIT MORPHISM HAS THE COMPLEMENTARY STRUCTURE MAPS AS KERNEL AND COKERNEL

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the kernel statement, giving both existence and uniqueness explicitly; the closing sentence 'By duality, (2) follows as did (1).' is the author's own routine duality remark on the part written out in full inside the counted span, and it is preserved verbatim.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

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Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Direct sums described by four morphisms. Same chapter, Remark remark-direct-sum. Supplies the identities satisfied by the two injections and two projections of a direct sum that identify the two structure maps as a kernel and a cokernel.

Lemma 0.1. *Let $\mathcal{A}$ be a preadditive category, x and y be objects of $\mathcal{A}$, and $j : y \to x$ and $q : x \to y$ be morphisms satisfying $q \circ j = id_y$. Then*

(1) *$id_x - j \circ q$ has kernel j.*
(2) *$id_x - j \circ q$ has cokernel q.*

Proof. As for (1), compute first that $(\mathrm{id}_x - j \circ q) \circ j = 0$. We must show that j satisfies the universal property of the kernel of $\mathrm{id}_x - j \circ q$, i.e., that given a morphism $a : z \to x$ satisfying $(\mathrm{id}_x - j \circ q) \circ a = 0$ there exists a unique morphism $b : z \to y$ such that $a = j \circ b$. Existence follows by computing that $a = j \circ q \circ a$ (so that we can take $b = q \circ a$). Uniqueness follows from that $b = q \circ j \circ b = q \circ a$ (so that a determines b).

By duality, (2) follows as did (1). □

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